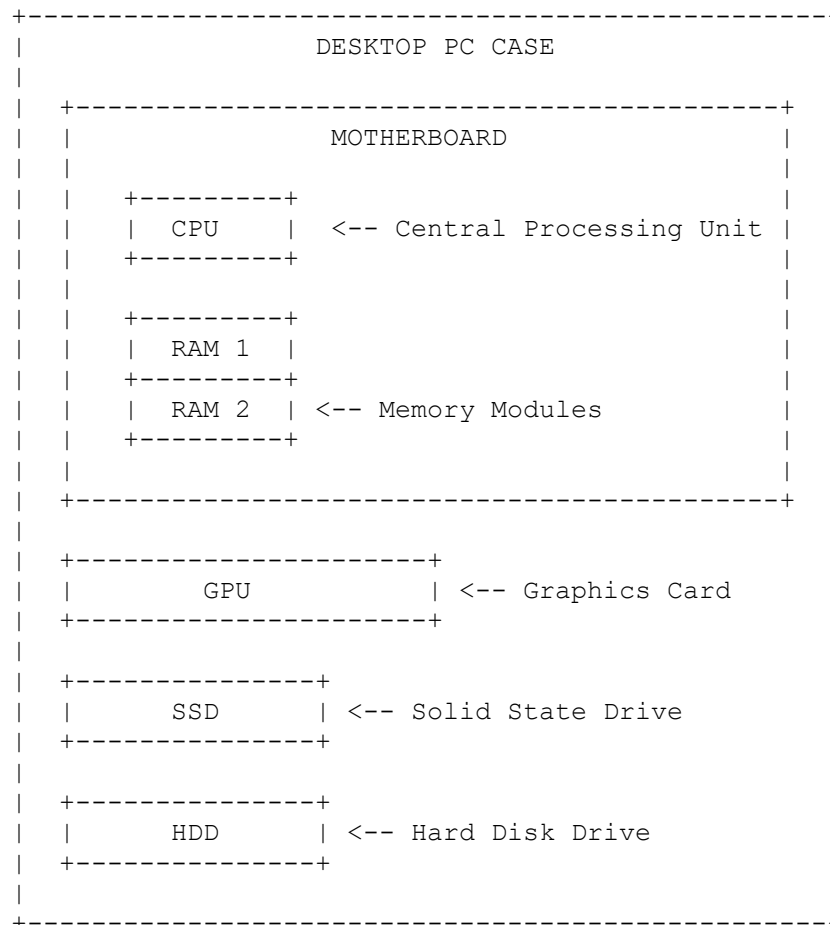


Overview of Hardware Components – Assignment

1. Labeled Diagram of a Desktop PC



2. HDD vs SSD Comparison Table

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower (50–150 MB/s)	Much faster (500 MB/s to several GB/s)
Durability	Contains moving parts, easier to damage from shocks	No moving parts, more shock-resistant
Noise	Makes some noise while operating	Silent operation
Power Consumption	Uses more power	Uses less power
Boot Time	Slower startup	Faster startup
Cost per GB	Cheaper	More expensive
Typical Use Cases	Large file storage, backups, budget PCs	Gaming PCs, laptops, operating systems, fast applications

Summary: SSDs are faster and more durable, while HDDs offer larger storage at a lower cost.

3. Role of CPU, RAM, and GPU in Games (PUBG/Free Fire)

CPU (Central Processing Unit)

The CPU acts as the brain of the computer. It processes game logic, player actions, AI behavior, and communication between different hardware components. A faster CPU helps games run more smoothly and reduces lag.

RAM (Random Access Memory)

RAM temporarily stores game data that the computer needs quickly. More RAM allows the game to load maps, textures, and player information efficiently. Insufficient RAM can cause stuttering and slow loading times.

GPU (Graphics Processing Unit)

The GPU is responsible for rendering graphics, animations, and visual effects. It determines how smoothly the game runs and how good the graphics look. A powerful GPU enables higher frame rates and better visual quality.

4. Hardware Upgrades for Running the Latest FIFA Smoothly

Component to Upgrade	Reason
GPU (Graphics Card)	Improves graphics quality, frame rates, and overall gaming performance.
CPU	Helps process game physics, AI, and background tasks more efficiently.
RAM	Upgrading to 16 GB or more prevents slowdowns and improves multitasking.
SSD	Reduces game loading times and speeds up system performance.
Power Supply (if needed)	Ensures enough power for upgraded components, especially a new GPU.
Cooling System	Keeps the CPU and GPU at safe temperatures during long gaming sessions.

Example Upgrade Plan

- Upgrade RAM from **8 GB to 16 GB**
- Replace HDD with a **512 GB or 1 TB SSD**
- Install a stronger **GPU** for better graphics performance
- Upgrade the **CPU** if it is outdated and bottlenecks the system

These upgrades would help FIFA run more smoothly, load faster, and maintain higher frame rates during gameplay.